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SWOT Analysis of Cloud Computing Environment

Sonal Dubey, Kritika Verma, M.A. Rizvi and Khaleel Ahmad

Abstract Cloud computing is a technology which deals with the collection of a large number of computers connected together on communication networks, for example the Internet. Cloud computing, dynamically increases computing capacity or add capabilities with minimum intervention of humans and without much investment in new infrastructure. At this moment cloud computing is at infancy stage, with many groups of providers, delivering cloud-based services, from full-scale applications to storage services. In the present era, IT sector turned into cloud-based services independently, but cloud computing integrators and aggregators as of now up and coming. The virtual servers in cloud computing virtually exist and so they can be scaled on the fly without affecting the client, to some extent in spite of being physically objectified, cloud gets expanded or compressed. In this paper, an attempt is made to do a SWOT analysis of a cloud computing environment as many users are in dilemma whether to use it or not. What are the benefits and disadvantages in using the cloud? A critical and detail analysis is done by mapping its Strengths (S), Weakness (W), Opportunity (O), and Threat (T) in different ways.

Keywords Cloud computing · SWOT · CSP · SLA · SWOT analysis

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1 Introduction

Cloud computing is an important improvement in the delivery of information and services. According to National Institute of Standards and Technology (NIST) definition, "*Cloud computing is a model for enabling ubiquitous, convenient, on-demand network access to a shared pool of configurable computing resources (e.g., networks, servers, storage, applications, and services) that can be rapidly provisioned and released with minimal management effort or service provider interaction*" [1].

There are three service models, five key characteristics, and four deployment models [2].

1.1 Key Characteristics of Cloud Computing

Self-Service On-Demand: It provides need-based services to end users who request resources at real time. The user can scale up and scale down the required infrastructure up to a desired level without disturbing the host job.
Broad Network Access: The capabilities available at the network can be accessed by the clients with the help of Internet.
Pooling of Resource: With the help of multitenant model, grouping of provider's computing resources is grouped with the help of different physical and virtual resources.
Rapid Elasticity: On-demand elastically scaling of delivering and releasing of capabilities (inward and outward) is done.
Service Measurement: With the help of abstraction suitable to the type of service the control and optimization of resource is done automatically with the use of metering ability [2].

1.2 Service Models of Cloud Computing

Software as a Service (SaaS): The provider's application that runs on the cloud infrastructure is used by the client. Various client devices such as thin client interface (e.g., web browser) or a program interface. Have accessibility to applications through the Internet.
Platform as a Service (PaaS): The end user deploys the applications that are made or procured on the cloud infrastructure, which are built up using programming languages, libraries, tool, and services supported by the provider.
Infrastructure as a Service (IaaS): The end user that provisions the processing, networks, storage, and other essential computing resources for the deployment and running of arbitrary software, which may include operating systems and applications [2].

1.3 Deployment Models of Cloud Computing

Private cloud: The cloud infrastructure is owned by a single organization that comprises of multiple clients for private use.
Community cloud: Different organizations with common concerns (e.g., mission, policy, security requirements and compliance considerations) own this cloud infrastructure and are managed by a specific community of consumers of such organization.
Public cloud: General public can openly use the cloud infrastructure.
Hybrid cloud: The cloud infrastructure can be classified in at least two distinctive cloud infrastructures (private, community, public) that represent idiosyncratic units, however, are bound together by standardized or proprietary technology that allows data and application portability [2].

2 SWOT Analysis

As defined by Albert Humphrey, "*A SWOT analysis is a strategic planning tool used to evaluate the Strengths, Weakness, Opportunities, and Threats involved in a project or in a business venture or in any other situation of an organization or individual requiring a decision in pursuit of an objective*".

2.1 SWOT Definition

SWOT is an abbreviation for Strengths, Weaknesses, Opportunities, and Threats. SWOT is a simple model that evaluates what an organization can and cannot do as well as its possible opportunities and threats. As per the definition, Strengths (S) and Weaknesses (W) are the internal factors over which organization has some measure of control. Furthermore, by definition, Opportunities (O) and Threats (T) are measured to be external factors over which organization has basically no control [3]. The SWOT framework is represented in Fig. 1.

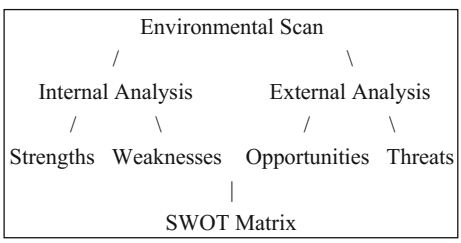


Fig. 1 SWOT analysis framework

The SWOT analysis includes the following segments as explained by Albert Humphrey.

- **Strengths** are usually internal features of an organization or business that gives it a benefit over its opponents. These are usually things like the quality, reputation, assets, people, value proposition, competitive advantages, geographical location, etc.
- **Weaknesses** are typically internal characteristics of an organization that place it as a weakness to the opponents. These are usually things like reputation or credibility, IT systems, financial aspects, people or business processes, loss of key staff, etc.
- **Opportunities** are external factors that present the opportunity to expand profits, sales, and/or get market share. This might include ingoing a marketplace wherein there may be excessive growth, developing new products, weak competition, improving marketing plans, or creating new partnerships.
- **Threats** are external factors which could present a hazard to the existing or future business. These threats may be outside the control of an organization. Examples of these threats may comprise price inflation or deflation, legal, financial conditions, competitive pricing or new competitors, or lack of key partners or contracts [4].

2.2 Need of SWOT Analysis of Cloud Computing

SWOT is a degree of analyzing a system to presentation its high-quality factors and awful factors for evaluation. Mostly, SWOT analysis is utilized to assess the state of affairs of marketplace when a person wants to enter in the market. Before moving into the cloud environment one needs to have some fare knowledge about the cloud and its services. Therefore, SWOT analysis gives few proofs; we are trying to assess the cloud computing on SWOT norm.

During our study of cloud computing we came across several opinions. Some of them were in favor of cloud computing while some were not, i.e., against cloud computing. Hence, due to the diversity in opinions about it, we tried to analyze it and thus we are doing a SWOT Analysis on cloud computing.

2.3 The SWOT Matrix

To flourish procedures taking into account the SWOT outline, a matrix of these aspects can be created. The TOWS matrix, also known as a SWOT Matrix is shown in Table 1.

Table 1 TOWS/SWOT matrix

	Strengths	Weaknesses
Opportunities	S-O strategies	W-O strategies
Threats	S-T strategies	W-T strategies

- **S-O approaches** practice opportunities that are appropriate to the organization's strengths.
- **W-O approaches** overcome weaknesses to follow opportunities.
- **S-T approaches** discover ways that an organization can use as its strengths to diminish its susceptibility to threats.
- **W-T approaches** set up a self-protective plot to thwart the organization's weaknesses from making it extremely vulnerable to threats.

2.4 Strengths of Cloud Computing

S1 Data sharing.
S2 Scalable storage space.
S3 Enhance availability of data and services.
S4 Development of OS-independent applications.
S5 Improved performances of services.
S6 Reliability of data and services.
S7 Reduced cost (pay-per usage).
S8 Work load diversity.
S9 Power-management flexibility.
S10 The cloud offerings are heterogeneous and without agreed interfaces.
S11 Increased security level for devices achieved by a centralized monitoring and maintenance of software.
S12 Number of new functionalities might be offered.
S13 Portability of application is possible.
S14 Enhance business agility while maintaining IT security and control.
S15 Attain cloud economics while leveraging existing IT investments.
S16 Pricing transparency.
S17 Controlled interface, ubiquitous access, location independence, sourcing autonomy, virtual professional environments, and hurried elasticity.
S18 Delivering variable costs, reduced capital expenditure, lower staff costs, cost efficient consolidation of physical resources.
S19 Simplifying access and management of applications and IT resources in a virtualized on-demand model.
S20 Flexible and resilient in disaster recovery.
S21 Service Level Agreement (SLA) guarantees the services from the cloud service providers to client.

2.5 Weakness of Cloud Computing

W1 Interoperability (communications between the cloud).
W2 Global compliance problem/different compliance in different regions.
W3 Open Standards.
W4 A bulk loads are limited by bandwidth.
W5 Cloud services are multi-vendors solutions by definition.
W6 Very few true hybrid cloud offerings exists today.
W7 Modification on maintenance model.
W8 Paradigm swing at present occupational and IT departments.
W9 No easy migration to another CSP (Cloud Service Provider).
W10 More and in depth knowledge is required for managing and implementing SLA contracts with CSP's.
W11 Third-party/CSP dependability for computing services.
W12 High-speed Internet connection essential to connect to cloud database.
W13 No precise method to select/find CSP.

2.6 Opportunities of Cloud Computing

O1 Overcoming latency limitations.
O2 Improving bandwidth utilization.
O3 Dynamic network monitoring.
O4 Technical issues resolution.
O5 CRM (Customer Relationship Management) has become one of the most accessible opportunity.
O6 Market entry or application deployment is cheaper, quicker repayment of improvement charges, and superior return on investment.
O7 Better comprehension of process and governance risk will transfer the preferences of IT owners toward the auditable and very professional security practices of CSPs.
O8 Customers will become more aggressive in dropping their cost of both business and personal computing, and will become far more accepting of lightweight client machines running free and open-source operating systems and applications.
O9 Overall growth in development demand will expand the significance of high-leverage application frameworks that allow quicker development of higher quality products.
O10 Adaptive to future needs.
O11 Cloud provides an excellent back-end for mobile applications.
O12 Optimizing the usefulness and proficiency of cloud computing environments.

O13 Mitigating identity, privacy, security, reliability, and manageability risks in cloud-based environments, as they vary from traditional data centers.
O14 Expansion and growth.
O15 The more effective use of computer resources to help the environment and encourage energy saving.
O16 Most of the cloud providers replicate user's data in multiple places. This multiplies redundancy of data and data independence from system failure offers a level of disaster recovery.
O17 CSP has the ability to relocate security resources for filtering, traffic shaping or encryption, dynamically, in order to increase the defensive measures.
O18 The organization can concentrate on serious tasks without having to experience additional costs with respect to IT staffing and training.
O19 The cloud computing approach speeds the deployment while preserving dynamic flexibility.

2.7 *Threats of Cloud Computing*

T1 Security issues (Privacy, Authentication, Malware, Confidentiality, Third party, Data integrity, Loss of data: Is foreseeable, thus backup for every usage has to be maintained on the PC).
T2 Management of data.
T3 Data ownership.
T4 Connectivity, Speed, Safety and Efficiency.
T5 Virtual security is critical and security concern must be addressed on the path to hybrid cloud deployments.
T6 Physical location of software and hardware is unidentified. Site investigations and inspections are hard.
T7 Quality problems with CSP.
T8 Measurement of useful resource utilization and end user activities lies in the hands of the CSP.
T9 Opaque cost structure because of greater flexible usage of cloud services.
T10 No or little insight in CSP uncertain procedures.
T11 Natural calamities.
T12 Migration from one cloud platform to another is difficult.
T13 Portability: porting of services/application at another location may be cumbersome.
T14 Vendor shutdown: the vendor may suddenly go out of business and close the shop. This would create a very unstable situation for consumer [5].

After going through a lot of data and analysis done in the above-mentioned Table 2, it is analyzed that to follow opportunities which are appropriate to the firm's strengths are:

Table 2 Mapping of strength with opportunities (S-O)

Strengths	Opportunities
S1	O1 + O2 + O11 + O19
S2	O9 + O12 + O14 + O19
S3	O1 + O2 + O11 + O16
S4	O9 + O10
S5	O1 + O2 + O3 + O11 + O13 + O17
S6	O13 + O16 + O17
S7	O6 + O8 + O18
S8	O3 + O16 + O19
S9	O12 + O15
S10	O9 + O10
S11	
S12	O9 + O17 + O19
S13	O3 + O11 + O13
S14	O5 + O6 + O7 + O8
S15	O6 + O8 + O18
S16	O5 + O7 + O8
S17	O1 + O2 + O3 + O10 + O11 + O12 + O13 + O16 + O17
S18	O6 + O18
S19	O3 + O19
S20	O16 + O17 + O19
S21	

- Security of the data will improve
- Data sharing with ease
- Scalability and availability is easier
- Business agility is increased
- Cost will be reduced
- Service Level Agreement guarantees the services
- Heterogeneous environment is enhanced.

It can be summarized that many opportunities will be open for the betterment for business and organizations with the arrival of a cloud computing environment.

After analyzing the above-mentioned data in Table 3, it is analyzed that when the cloud will establish itself in the marked and advancement done in the cloud technology, weakness will be overcome automatically there will be opportunities only with cloud. The future of the Internet is cloud only. The opportunities of the cloud are

- Communication between the clouds is established
- Global fixed standard is introduced
- Bandwidth is improved
- Maintenance model is enhanced.

Table 3 Mapping of weakness and opportunities (W-O)

Weakness	Opportunities
W1 + W6 + W9	O3 + O5 + O9 + O11 + O12 + O14 + O19
W2 + W3 + W5	O5 + O7 + O8 + O9 + O13 + O14 + O19
W4	O1 + O2 + O3 + O9 + O10 + O12 + O14 + O19
W7	O7 + O10 + O12 + O13 + O17
W8	O6 + O8 + O9 + O12 + O14 + O15 + O19
W10 + W11	O5

It can be summarized that many opportunities will be open for the betterment of a cloud computing environment.

By analyzing the mentioned Strengths and Threats in Table 4, it is analyzed that to reduce the vulnerabilities to external threats we have to concentrate on some strengths of cloud like

- Security of the cloud should be improved
- Data sharing, scalability, and availability should be simplified
- Business agility should be increased
- Cost should be reduced
- SLA guarantees the services
- Heterogeneous environment should be enhanced.

It can be summarized that improving all the above mentioned threats in turn cloud computing environment will more popular.

By analyzing the above-mentioned Weakness with Threats in Table 5, it can be said that to establish a self-protective plan to thwart the organization's weaknesses from making it highly susceptible to external threats are listed below

Table 4 Mapping of strength with threats (S-T)

Strengths	Threats
S5 + S6 + S8 + S11 + S14 + S21	T1
S1 + S2 + S3 + S19 + S20	T2
S14 + S17 + S21	T3
S2 + S4 + S8 + S17	T4
S6 + S14 + S15 + S17	T5
S5 + S17 + S18 + S20	T7
S21	T8
S7 + S15 + S16 + S18	T9
S21	T10
S6 + S13 + S20	T11
S10 + S13	T12 + T13
S21	T14

Table 5 Mapping of weakness with threats (W-T)

Weaknesses	Threats
W1 + W6 + W9	T1 + T4 + T12
W2 + W3 + W5	T2 + T3 + T9 + T12 + T13
W4	T2 + T4 + T7 + T13
W7	T1 + T2 + T4 + T5 + T7 + T12 + T13
W10 + W11 + W13	T1 + T2 + T3 + T5 + T7
W12	T2 + T4 + T7 + T12

- Communication between the clouds should be established
- Global fixed standard should be introduced
- Bandwidth should be improved
- Maintenance model should be enhanced.

It can be summarized that then most of the threats will be vanished for the betterment of a cloud computing environment.

3 Conclusion

After above-mentioned marathon analysis and discussion on Strength (S), Weakness (W), Opportunity (O), and Threat (T) of cloud computing. It can be concluded that cloud computing has yet not reached a maturity that leads it into a fruitful stage. However the majority of the key problems with cloud computing have been solved to a certain point that Cloud computing has become very popular for commercial and organizational utilization. This does not mean that all the problems listed above have been completely resolved, only that the accordingly risks can be accepted to a certain point. Cloud computing is thus still a very popular area for research. It can be summarized that although there are few threats and weaknesses in cloud computing, but on the contrary, there are various strengths and opportunities thus this technology is getting popular and the future is cloud. This technology will get better day by day as researches will take place.

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